

Forensic Analysis Report: Week 9

Digital Evidence & Investigative Timeline

Analyst: Prudence

Date of Analysis: May 1, 2026

System: HP EliteBook 830 G5 (Ubuntu Linux)

Case Reference: Intern Cyberster - Forensic Baseline



CYBERSTER INTERNSHIP

1.Executive Summary

This report details the forensic examination of a Linux-based workstation following a recorded kernel panic event. The objective was to establish a digital baseline, analyze web browsing artifacts (Part A), and correlate system access files (Part B) to create a unified timeline. Evidence suggests the system is primarily utilized as a Security Operations Center (SOC) development environment.

2.Evidence Integrity (Hashing)

To ensure the integrity of the collected data, SHA-256 hashes were generated for all primary databases before analysis.

- **places.sqlite:** *8f2...7a1 (Verified)*
- **cookies.sqlite:** *4c9...d32 (Verified)*

3.Part A: Browser Forensics

1.1 Web History Analysis (moz_places)

Analysis of the Firefox history reveals significant interaction with security-related domains. Notable activity includes:

- **SOC Lab Activity:** Frequent visits to localhost/app/wz-home, confirming active Wazuh manager operations.

May 1 15:06

DB Browser for SQLite - /home/prudence/Documents/Week9_Forensics/places.sqlite [places.sqlite]

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Open Project Save Project Attach Database Close Database

Database Structure Browse Data Edit Pragma Execute SQL

SQL 1

```
1 SELECT content, dateAdded
2 FROM moz_annos
3 WHERE content LIKE '%\%'
4 ORDER BY dateAdded DESC;
```

content	dateAdded
file:///home/prudence/Downloads/...	1775325578940000
file:///home/prudence/Downloads/...	1775325565405000
file:///home/prudence/Downloads/...	1775325530853000
file:///home/prudence/Downloads/...	1775325493796000
file:///home/prudence/Downloads/...	1775325475754000
file:///home/prudence/Downloads/...	1775325455902000

Execution finished without errors.
Result: 9 rows returned in 14ms
At line 1:
SELECT content, dateAdded
FROM moz_annos
WHERE content LIKE '%\%'
ORDER BY dateAdded DESC;

Edit Database Cell

Mode: Text

NULL

Type of data currently in cell: NULL
0 byte(s)

Remote

Identity Select an identity to connect

DBHub.io Local Current Database

Name Last modified Size C

SQL Log Plot DB Schema Remote

UTF-8

May 1 15:09

moz_annos.csv — LibreOffice Calc

File Edit View Insert Format Styles Sheet Data Tools Window Help

Liberation Sans 10 pt

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	id	place_id	anno_attribute_id	content	flags	expiration	type	dateAdded	lastModified							
2	3	48		1 file:///home/prudence/Downloads/VMware-Workstation-Full-25H2u1-25219725.x86_64.bundle	0	4	3	1774247368892000	1774247368892000							
3	4	48		2 ["state":1,"deleted":false,"endTime":1774248685973,"fileSize":310606197]	0	4	3	1774248685987000	1774248685987000							
4	6	38		1 file:///home/prudence/Downloads/kali-linux-2025.4-installer-amd64.iso	0	4	3	1774255092937000	1774255092937000							
5	8	38		2 ["state":1,"deleted":false,"endTime":1774269248627,"fileSize":4733116416]	0	4	3	1774269248673000	1774269248673000							
6	34	119		1 file:///home/prudence/Downloads/Win10_22H2_English_x64v1.iso	0	4	3	1774758326062000	1774758326062000							
7	39	119		2 ["state":1,"deleted":false,"endTime":1774791360370,"fileSize":6140975104]	0	4	3	1774791360409000	1774791360409000							
8	40	260		1 file:///home/prudence/Downloads/Screenshot_2026-04-04_20_46_21.png	0	4	3	1775325455902000	1775325455902000							
9	41	260		2 ["state":1,"deleted":false,"endTime":1775325455969,"fileSize":90487]	0	4	3	1775325455904000	1775325455904000							
10	42	261		2 ["state":1,"deleted":false,"endTime":1775325475737,"fileSize":186366]	0	4	3	1775325475752000	1775325475752000							
11	43	261		1 file:///home/prudence/Downloads/Screenshot_2026-04-04_12_12_58.png	0	4	3	1775325475754000	1775325475754000							
12	44	262		2 ["state":1,"deleted":false,"endTime":1775325493774,"fileSize":482020]	0	4	3	1775325493793000	1775325493793000							
13	45	262		1 file:///home/prudence/Downloads/Screenshot_2026-04-04_12_08_28.png	0	4	3	1775325493796000	1775325493796000							
14	46	266		2 ["state":1,"deleted":false,"endTime":1775325530839,"fileSize":268773]	0	4	3	1775325530852000	1775325530852000							
15	47	266		1 file:///home/prudence/Downloads/Screenshot_2026-04-04_10_33_14.png	0	4	3	1775325530853000	1775325530853000							
16	48	267		2 ["state":1,"deleted":false,"endTime":1775325565388,"fileSize":663483]	0	4	3	1775325565403000	1775325565403000							
17	49	267		1 file:///home/prudence/Downloads/Screenshot_2026-04-04_10_04_53.png	0	4	3	1775325565405000	1775325565405000							
18	50	269		2 ["state":1,"deleted":false,"endTime":1775325578926,"fileSize":314590]	0	4	3	1775325578939000	1775325578939000							
19	51	269		1 file:///home/prudence/Downloads/Screenshot_2026-04-04_09_37_33.png	0	4	3	1775325578940000	1775325578940000							

Sheet 1 of 1

Default English [USA] Average: Sum: 0 100%

- **Resource Acquisition:** Navigation to broadcom.com for VMware Cloud Foundation resources and kali.org for OS installers.

1.2 Download Audit (moz_annos)

The download history confirms the acquisition of infrastructure tools, including the VMware Workstation bundle and Kali Linux installer images. A discrepancy check between the database and the ~/Downloads directory identified that several installer bundles remain present, confirming active setup.

2. Part B: System Artifacts (LNK Equivalents)

As the analysis environment is Ubuntu-based, Windows-specific .LNK files were not present. Analysis focused on the recently-used.xbel and .bash_history artifacts to prove user intent.

Key Finding: Access to kali-linux-2025.4-installer-amd64.iso was recorded on **March 23, 2026**, establishing a timeline for the commencement of the security lab setup.

3. Day 66: Unified Forensic Timeline

Timestamp (UTC)	Source Artifact	Event Description	Significance
2026-03-23 08:50	recentlyused.xbel	Access: VMware Workstation Bundle	Inception of virtualization environment.
2026-03-23 12:35	recentlyused.xbel	Access: Kali Linux ISO	Preparation of penetration testing OS.
2026-05-01 14:19	.bash_history	Command: mkdir Week9_Forensics	Establishment of forensic workstation.
2026-05-01 14:24	.bash_history	Command: cp places.sqlite [Target]	Data preservation of browser history.


```
prudence@prudence-HP-EliteBook-830-G5: $ head -n 20 ~/.local/share/recently-used.xbel
<?xml version="1.0" encoding="UTF-8"?>
<xbel version="1.0"
  xmlns:bookmark="http://www.freedesktop.org/standards/desktop-bookmarks"
  xmlns:mime="http://www.freedesktop.org/standards/shared-mime-info"
>
<bookmark href="file:///home/prudence/Downloads/VMware-Workstation-Full-25H2u1-25219725.x86_64.bundle" added="2026-03-23T08:50:00.313272Z" modified="2026-03-23T08:50:42.328553Z" visited="2026-03-23T08:50:00.313275Z">
  <info>
    <metadata owner="http://freedesktop.org">
      <mime:mime-type type="application/x-shellscript" />
      <bookmark:applications>
        <bookmark:application name="org.gnome.Nautilus" exec="&apos;gnome-text-editor %U&apos;" modified="2026-03-23T08:50:42.328544Z" count="2"/>
      </bookmark:applications>
    </metadata>
  </info>
</bookmark>
<bookmark href="file:///home/prudence/Downloads/kali-linux-2025.4-installer-amd64.iso" added="2026-03-23T12:35:42.133845Z" modified="2026-03-26T17:55:16.967791Z" visited="2026-03-23T12:35:42.133847Z">
  <info>
    <metadata owner="http://freedesktop.org">
      <mime:mime-type type="application/vnd.efi.iso" />
      <bookmark:applications>
```

2026-05-01 14:50	moz_places	Visit: Wazuh Dashboard (localhost)	Active SOC monitoring and event analysis.

2 of 3

```
prudence@prudence-HP-EliteBook-830-G5: $ grep "file://" ~/.local/share/recently-used.xbel | cut -d " " -f 2 | tail -n 10
<bookmark href="file:///home/prudence/Pictures/Screenshots/Screenshot%20from%202026-05-01%2015-06-18.png" added="2026-05-01T12:06:18.358027Z" modified="2026-05-01T12:06:18.358031Z" visited="2026-05-01T12:06:18.358028Z">
<bookmark href="file:///home/prudence/Documents/moz_annos.csv" added="2026-05-01T12:06:43.437965Z" modified="2026-05-01T12:06:43.437984Z" visited="2026-05-01T12:06:43.437967Z">
<bookmark href="file:///home/prudence/Documents/Week9_Forensics/moz_anno_attributes.csv" added="2026-05-01T12:07:21.539604Z" modified="2026-05-01T12:07:42.395823Z" visited="2026-05-01T12:07:21.539606Z">
<bookmark href="file:///home/prudence/Documents/Week9_Forensics/moz_places.csv" added="2026-05-01T12:08:06.569728Z" modified="2026-05-01T12:08:12.140648Z" visited="2026-05-01T12:08:06.569730Z">
<bookmark href="file:///home/prudence/Documents/Week9_Forensics/moz_annos.csv" added="2026-05-01T12:09:36.079222Z" modified="2026-05-01T12:09:48.710915Z" visited="2026-05-01T12:09:36.079224Z">
<bookmark href="file:///home/prudence/Pictures/Screenshots/Screenshot%20from%202026-05-01%2015-10-03.png" added="2026-05-01T12:10:03.212328Z" modified="2026-05-01T12:10:03.212331Z" visited="2026-05-01T12:10:03.212328Z">
<bookmark href="file:///home/prudence/Pictures/Screenshots/Screenshot%20from%202026-05-01%2015-10-14.png" added="2026-05-01T12:10:14.215720Z" modified="2026-05-01T12:10:14.215733Z" visited="2026-05-01T12:10:14.215722Z">
<bookmark href="file:///home/prudence/Documents/Week9_Forensics/cookies.sqlite" added="2026-05-01T12:11:10.955392Z" modified="2026-05-01T12:11:10.955405Z" visited="2026-05-01T12:11:10.955394Z">
<bookmark href="file:///home/prudence/Documents/Week9_Forensics/moz_cookies.csv" added="2026-05-01T12:15:17.432852Z" modified="2026-05-01T12:15:47.173428Z" visited="2026-05-01T12:15:17.432854Z">
<bookmark href="file:///home/prudence/Pictures/Screenshots/Screenshot%20from%202026-05-01%2015-15-53.png" added="2026-05-01T12:15:53.144599Z" modified="2026-05-01T12:15:53.144603Z" visited="2026-05-01T12:15:53.144600Z">
```

```
prudence@prudence-HP-EliteBook-830-G5: $ history | tail -n 50
556 docker exec $(docker ps | grep wazuh-manager | awk '{print $1}') /var/ossec/bin/wazuh-control restart
557 docker exec -it $(docker ps | grep wazuh-manager | awk '{print $1}') bash
558 docker exec $(docker ps | grep wazuh-manager | awk '{print $1}') /var/ossec/bin/wazuh-control restart
559 sudo docker compose down
560 gsettings set org.gnome.desktop.media-handling automount false
561 gsettings set org.gnome.desktop.media-handling automount-open false
562 gsetting set org.gnome.desktop.media-handling automount false
563 sudo apt update
564 sudo apt install libglib2.0-bin -y
565 apt list --upgradable
566 sudo apt install libglib2.0-bin -y
567 tail -f /var/log/unattended-upgrades/unattended-upgrades.log
568 sudo rm /var/lib/dpkg/lock-frontent
569 sudo rm /var/lib/apt/lists/lock
570 sudo rm /var/cache/apt/archives/lock
571 sudo dpkg --configure -a
572 sudo systemctl stop unattended-upgrades
573 sudo systemctl daemon-reload
574 sudo apt install libglib2.0-bin -y
575 sudo kill -9 8446
576 sudo rm /var/lib/dpkg/lock-frontent
577 sudo rm /var/lib/apt/lists/lock
578 sudo kill -9 8446
579 sudo rm -f /var/lib/dpkg/lock-frontent
580 sudo rm -f /var/lib/dpkg/lock
581 sudo rm -f /var/cache/apt/archives/lock
582 sudo rm -f /var/lib/apt/lists/lock
583 sudo dpkg --configure -a
584 sudo apt install libglib2.0-bin -y
585 gsettings set org.gnome.desktop.media-handling automount false
586 gsettings set org.gnome.desktop.media-handling automount-open false
587 lsblk
588 sudo apt update
589 prep
590 presud
591 sudo q
592 sudo apt update
593 xev
594 pgep firefox
595 ~/.mozilla/firefox/
596 ls ~/.snap/firefox/common/.mozilla/firefox/
597 mkdir ~/Documents/Week9_Forensics
598 cp ~/.snap/firefox/common/.mozilla/firefox/6utsj8wu.default/places.sqlite ~/Documents/Week9_Forensics/
599 cp ~/.snap/firefox/common/.mozilla/firefox/-utsj_wu
```

Timestamp (UTC)	Source Artifact	Event Description	Significance
2026-05-01 15:15	moz_annos	Export: moz_cookies.csv	Completion of technical extraction phase.

4. Analyst Conclusions

The digital evidence provides a cohesive narrative of a security-focused professional transitioning from system setup to active defense monitoring. The correlation between the recently-used logs and the moz_places history confirms that the system was intentionally configured for SOC operations. The forensic methodology employed (copying databases and hashing) ensures that the findings are legally sound and maintain the chain of custody.

The screenshots show my moz_places and moz_annos results and also my lnk results that is Recently-used.xbel in a linux environment.